# Similar-Part Search: Choosing a Configuration

Step 4 of the CAD-to-AI mini pipeline | 187 parts from the ABC dataset plus one hand-modeled bracket

**In short**

- The search finds parts that *look* alike by comparing rendered images of them.
- Three design choices give **12 configurations**. Each was scored on tests where the right answer is known in advance.
- **Aligning each part before rendering is what matters most.** It raised the hit rate on randomly rotated parts from 23-70% to 100%.
- **Chosen:** canonical pose + DINOv2, stored in Qdrant both as a single averaged vector and as six per-view vectors.

## 1. What was compared

Every part is rendered from six fixed camera angles (front, back, left, right, top and isometric). An image model turns each picture into a vector, and parts whose vectors are close count as similar. Three choices shape how well that works:

| Choice | Options | What it means |
|---|---|---|
| **Pose** | world<br>canonical | *World* renders the part exactly as it was modeled. *Canonical* first turns it onto its own axes, found from its flat faces: longest side along X, shortest along Z. |
| **Model** | DINOv2<br>CLIP | DINOv2 learned from images alone, so it encodes what things *look like*. CLIP learned from images paired with captions, so it encodes what things are *called*. |
| **Pooling** | mean<br>max<br>MaxSim | How six views become one score. *Mean* and *max* merge the six vectors into one. *MaxSim* keeps all six, and each view of the query looks for its best-matching view in the candidate. |

2 poses × 2 models × 3 poolings = 12 configurations.

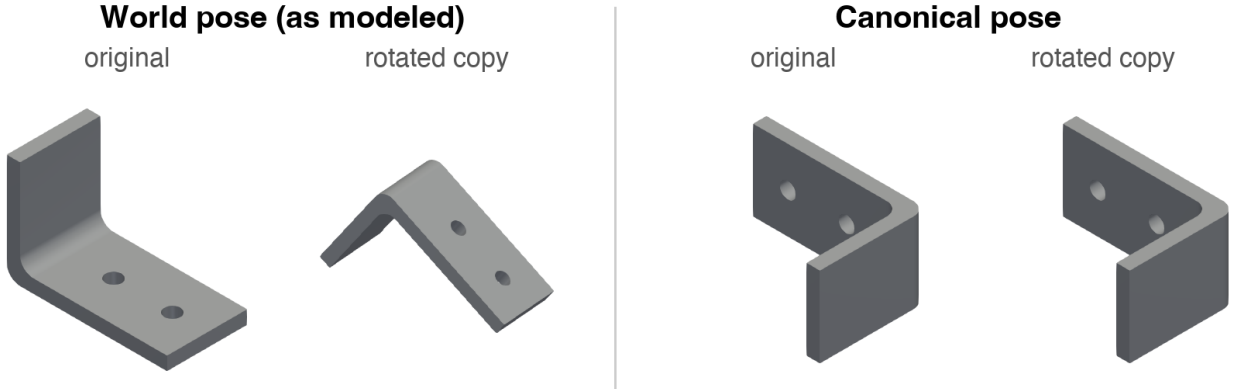



Figure 1. The bracket and a randomly rotated copy of it. In world pose the camera sees two different pictures. Canonical pose turns both back to the same position, so the pictures match.

## 2. How it was tested

Each test takes a part, changes it in a known way, and searches with the changed version. The search succeeds if the original part comes back as the **#1 result** (top-1).

- **Axis swap** (60 parts): the part is turned in 90° steps. This is the realistic case: the same part modeled lying a different way, e.g. its long side along Y instead of X.
- **Random rotation** (60 parts): the part is turned to any 3D angle. This is the worst case.
- **Duplicate** (43 copies): identical copies of a part inside the same assembly file. Most copies were only moved, not turned, so this is a sanity check rather than a real challenge.

## 3. Results

| Pose | Model | Pooling | Axis swap | Random rotation | Duplicate |
|---|---|---|---|---|---|
| **world** | DINOv2 | mean | 80% | 37% | 100% |
| | | max | 85% | 50% | 93% |
| | | MaxSim | 87% | 70% | 93% |
| | CLIP | mean | 72% | 23% | 95% |
| | | max | 75% | 27% | 93% |
| | | MaxSim | 78% | 53% | 93% |
| **canonical** | DINOv2 | mean | 100% | 100% | 98% |
| | | max | 100% | 100% | 98% |
| | | MaxSim | 100% | 100% | 98% |
| | CLIP | mean | 100% | 100% | 98% |
| | | max | 100% | 100% | 95% |
| | | MaxSim | 100% | 100% | 95% |

Table 1. Share of tests where the original part came back as the #1 result. Duplicates come back in the top 5 for every configuration.

### What the table shows

- **Alignment is the biggest lever.** Without it, a randomly rotated part finds itself only 23-70% of the time. With canonical pose, every configuration reaches 100%.
- **Without alignment, the expected order holds:** DINOv2 beats CLIP in every row, and MaxSim beats max, which beats mean. Smarter matching recovers part of the loss (DINOv2: 37% to 70% on random rotations), but not all of it.
- **With alignment, this test can no longer separate the configurations.** Model and pooling had to be chosen another way (section 4).
- **Duplicates miss #1 now and then** because the database holds near-identical "twin" parts. The copy ties with a twin and lands at #2.

## 4. Choosing the model by looking at results

With the rotation test maxed out, DINOv2 and CLIP were compared on real queries: canonical pose, MaxSim, top 5 results each. The two models agree on 3 of the top 5 results on average.

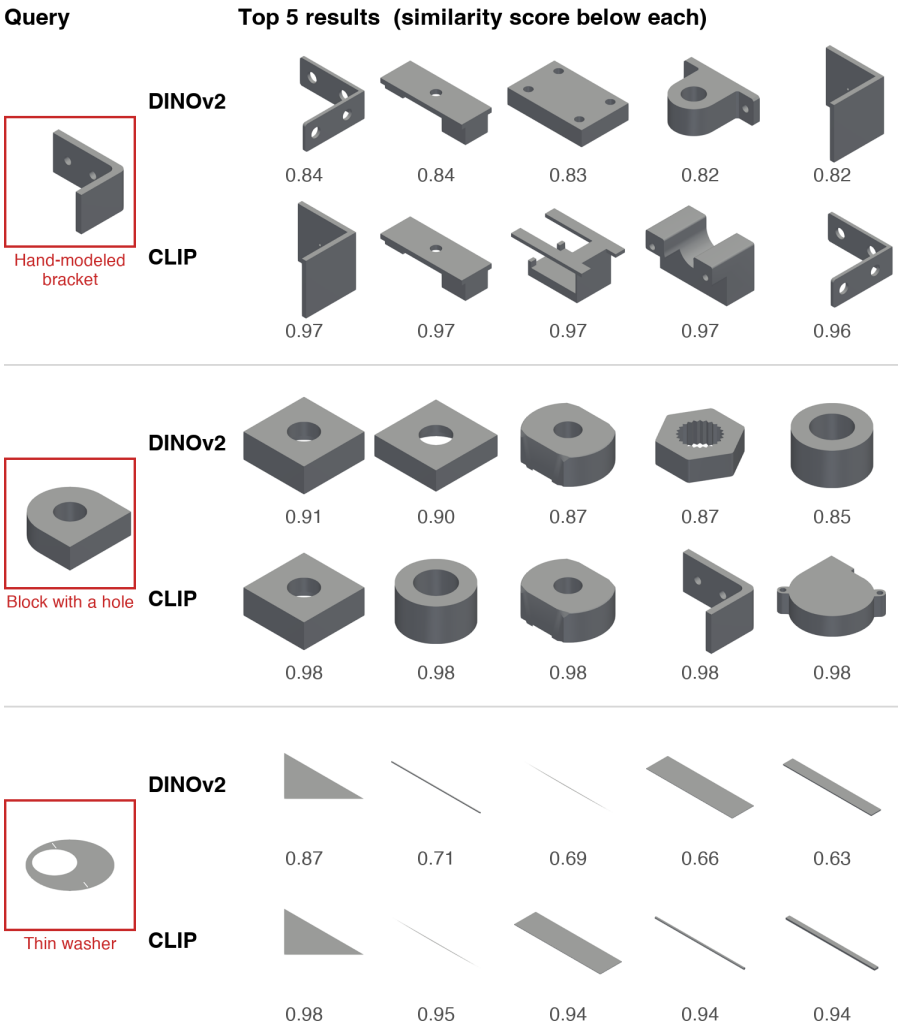



Figure 2. Three queries (red) and each model's top 5 results, with similarity scores. Isometric views shown.

- **Bracket:** both models return L-brackets and plates with holes, which are sensible matches.
- **Block with a hole:** DINOv2's five results are all blocks or rings with a hole. CLIP slips in an unrelated bracket at #4.
- **Thin washer:** both fail because there is no other washer in the data. But DINOv2's score drops from 0.87 to 0.63 down its list, while CLIP still rates everything 0.94 or higher.

That last point decided it. DINOv2's scores spread from about 0.6 to 0.9, so a cutoff for "nothing similar found" can actually work. CLIP squeezes almost everything into 0.90-0.98.

## 5. Decision

| | | |
|---|---|---|
| **Pose** | **Canonical** | Turns rotation from the main failure into a solved problem. |
| **Model** | **DINOv2** | Better matches in the manual check, and a score spread that supports a cutoff. |
| **Pooling** | **MaxSim (default) + mean** | Both are stored, since they cost little at this scale. MaxSim copes better when a symmetric part lands in an equivalent but different pose. Mean is a single vector, so it is the cheaper option at larger scale. |
| **Storage** | **Qdrant, local** | 187 parts. Each holds both vectors plus size and face counts, for filtering later. |

## 6. Known limitations and next steps

- **The search ignores size.** Every render is scaled to fill the frame, so the top match for the 60 mm bracket is a 20 mm part. Next step: filter or re-rank results by bounding-box size.
- **The tests no longer separate good from better.** A fair comparison of models needs pairs of parts that people have labeled as similar.
- **No "nothing similar" answer yet.** The washer query still returns five results. DINOv2's score spread makes a cutoff possible, but it still has to be set.
- **Canonical pose fails on one part out of 187.** Symmetric parts can also land in equivalent poses. MaxSim covers that case.